

Recording

Date	:	17/06/2026
Problem Specification	:	Find the specific number index and sorting the array
Assumption	:	The user enter the valid integers for array
Limitation	:	letters cant input
Input	:	Array size,Array elements, Number that have to search index
Processing	:	Read array Read the number to search Find the index of given number Sort the array
Output	:	Index of the searched number Sorted array in ascending order
Algorithm	:	Step 1: Read the array size n. Step 2: Create an integer array of size n. Step 3: Read n elements into the array. Step 4: Read the search value. Step 5: Search the array sequentially. Step 6: If the value is found, return its index. Step 7: If not found, return -1. Step 8: Sort the array using Bubble Sort. Step 9: Display the sorted array. Step 10: End the program.
Programme listing	:	Programme file attached
Test data and expected output	:	1. Test data: Size = 5 Array = 30 10 50 20 40 Search = 20 Expected output: Index = 3 Sorted Array = 10 20 30 40 50 2. Test data: Size = 4 Array = 5 8 2 1 Search = 7

Expected output: Index = -1

Sorted Array = 1 2 5 8

Output obtained for test data :

1. Test data: Size = 5
 Array = 30 10 50 20 40
 Search = 20
 Obtained output: Index = 3
 Sorted Array = 10 20 30 40 50

2. Test data: Size = 4
 Array = 5 8 2 1
 Search = 7
 Obtained output: Index = -1
 Sorted Array = 1 2 5 8

Analysis : The numbers of operation required in performing the algorithm.

	+, -	/, *	%	</>/<=>=
For calculation	N		-	N
Do while loop	-	-	-	-

Conclusion : The program successfully work. sorts the array in ascending order using Bubble Sort

Discussion : If the searched number is not found, the program returns **-1**.